

3	(a) same temperature no (net) transfer of thermal energy (between the bodies)	B1	[2]
	(b) (i) 41.3 K	B1	[1]
	(ii) 330.4 K	B1	[1]

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$$\begin{aligned} \text{(c)} \quad \Delta E_K &= \frac{3}{2} \times 1.9 \times 60 \\ &= 171 \text{ J} \end{aligned}$$

C1

$$\begin{aligned} \text{work done} &= p\Delta V \\ &= 1.2 \times 10^5 \times 950 \times 10^{-6} \\ &= 114 \text{ J} \end{aligned}$$

C1

C1

$$\begin{aligned} \text{thermal energy} &= 114 + 171 \\ &= 285 \text{ (290) J} \end{aligned}$$

A1 [4]